Interface Service Manager

Installation and Configuration Guide

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SECTION 1

Overview of the Interface Service Manager

Interface Service Manager (ISM) manages Hexagon services, including interfaces and core services. ISM provides features beyond what Microsoft Windows provides, including support for failing over a service to another computer. ISM also provides a convenient user interface for managing the Intergraph Public Safety interfaces and services.

For more information on how ISM relates to the overall HxGN OnCall® Dispatch system, see the *HxGN OnCall Dispatch Installation and Configuration Guide*.

The ability to start a service automatically at runtime and to stop the service at system shutdown or on demand are basic features provided by a Microsoft® Windows service.

ISM addresses the following two failures:

- Service failure - A service failure is the failure (exit) of a given instance of an service. ISM detects when a service terminates abnormally and attempts to re-spawn a new instance of that interface in its place. ISM assumes that if a service has not exited, the interface can perform its duties.
- Hardware failure - In this scenario, a node has failed. The ISM peer requests all interfaces to proceed to the primary state.

ISM resolves both types of failures. Each ISM instance communicates with its peer and coordinates the transfer of control between the interfaces and services executing on each server. Interfaces that maintain state information must support supplemental processing while in a secondary and primary state to avoid the loss of information during transition.

! ISM has runtime dependencies on RabbitMQ and the HxGN OnCall Dispatch database.

Features

The features of ISM include the following:

- Spawning each necessary service at run-time.
- Spawning multiple copies of a single executable as necessary.
- Spawning the processes in some necessary ordering.
- Re-spawning an executable upon premature termination.
- Terminating a service without stopping the ISM.

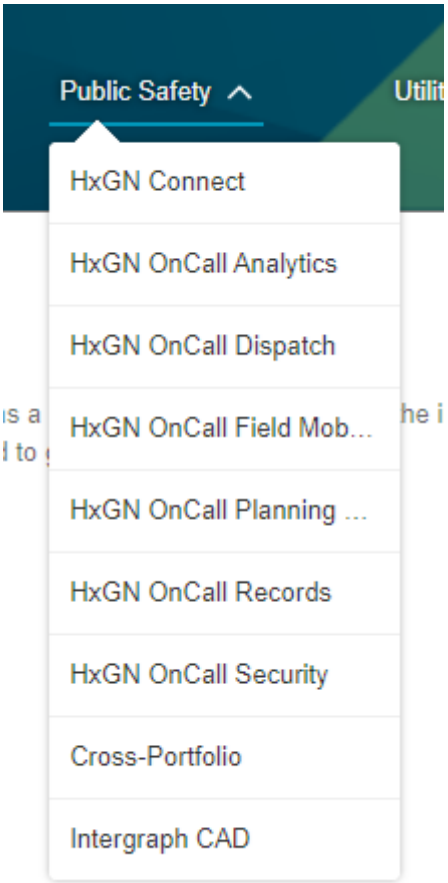
- Defining, starting, and/or removing a new interface or service without having to stop ISM.
- Providing cleanup upon directed termination through STOP or SHUTDOWN.
- Coordinating with a remote ISM peer to decide whether to respawn an aborted interface locally or request its peer to start a replacement. ISM only supports a one-to-one relationship with its peer.
- Managing ISM remotely.
- Storing the composite Service history locally in *\Intergraph\CAD\ISM*.
- Providing access to multiple ISM service instances through ISM Administration.
- Refreshing applications by monitoring the *\Intergraph\CAD\ISM* folder for changes, meaning you can add, update, and delete application instances without stopping the ISM service.

What's New

To read about new features, updated features, and system requirements, use the **Support** page. For online and phone support, see *Technical Support* (on page 20).

1. Go to the *Support page* *https://hexagon.com/support-success/safety-infrastructure-geospatial*.
2. Click **Log in**.
3. Type your username and password, and click **Log in**.

4. From the **Products** list, select your product portfolio.



The list of products for the selected portfolio displays.

5. Click a product link to open its support page.

6. When the product page opens, click the appropriate links to view product announcements or technical documentation.
   - Click **Release Notes** to review new or enhanced product features.
   - Click **Issues Resolved** to review defects that have been fixed.
   - Click **Supported Environments** to review system requirements.

- Click **Open-Source Software Notices** in the **Product Information** box, if available, to review information on open-source software in the product.

Some minor releases might not provide Release Notes or Issues Resolved.

Installing Interface Service Manager

Hexagon recommends running the HxGN OnCall Dispatch services under the ISM service. You need ISM on the application server(s) where you plan to run OnCall Dispatch services, such as Core Services, Recommend Unit, Scheduler service, Calendar service, LOI service, Change Publisher service, Geofence service, Tracker, and the Rest API. ISM must be installed on the web servers where you plan to run OnCall Dispatch web applications, including the OnCall Dispatch Website, WebApi, RealTime WebApi, and Identity.

After installing the backend services, you must configure them to run under ISM.

If you install the HxGN OnCall Dispatch applications using the *install_all_web_apps.bat* and *install_all_services.bat* files, those batch files install ISM. If you are installing applications on servers without using those batch files, then you will need to install ISM separately.

To install ISM:

1. Open Command Prompt using the **Run as administrator** option.
2. Navigate to the *\Services* folder in your installation media.
3. Run the *install_ism.bat* batch file. The scripts display output on the Command Prompt window.

Installation Notes

- Applications that use ISM must be configured by using the ISM Administration Tool or by creating a JSON configuration file and copying it to *C:\ProgramData\Intergraph\Cad\ISM*. See the *Interface Service Manager Help* for more information on using the Administration Tool.
- The ISM installation verifies that the *\Intergraph\Cad\cad-global.json.config* file exists and that its MESSAGE_BUS_CONN_STRING and MESSAGE_BUS_PASSWORD fields are defined.
- In on-premise installations, the *cad-global.json.config* file is located on the same computer from which ISM and ISM Administrator are run. If HxGN OnCall Dispatch is running in the cloud and ISM and ISM Administrator are running on-premise, RabbitMQ must also be installed on-premise.
- When ISM is installed or upgraded using choco commands, the ISM service account is set to Local System by default. The *SetISMUser.ps1* file (located at

*ISM**Scripts*) can be used to set the ISM service account to the domain account. For more information and examples, run the `get-help .\SetISMUser.ps1` command from a PowerShell window.

Set Up Peering in ISM

By default, OnCall Dispatch does *not* require peering.

If you have add-ons that require running in a peered configuration, perform the following steps.

When running applications in a peered setup, ISM must be configured on both servers so that ISM can detect which system is its peer. Because peering requires sending packets between the peered OnCall Dispatch instances and because increased network traffic can impact overall performance, Hexagon recommends using peering only when necessary.

To set up two computers to be peered in ISM:

1. Run ISM Administrator.
2. From the menu, select **Manager > Peer Settings**.
3. The default peer wait delay is 10 seconds. Take the default.
4. Specify the peer name. This is the name of the peered system.
5. Click **Apply**.

Localize ISM Administrator

The language used to run ISM Administrator can be configured to meet the needs of your site. For more information on configuring language options, see the *ISM Administrator Localization* (on page 17) topic.

Configuring ISM

Setting Up Interface Service Manager

By default, ISM is installed and runs under the Local System Account. This means ISM and the child processes (applications) execute with administrator-like rights.

The default method of running ISM permits the application to interact with the desktop to diagnose application issues. When ISM is running using the Local System Account, ISM does not support ISM Administrator's **Run As** application settings.

The other ISM user scenarios are a *domain administrator account* or a *domain user account* (see "*Configuring ISM for a Domain User Account*" on page 12).

- With a domain administrator account, the account is a member of the local administrators and must have the "Logon as a service" privilege, which is granted to the user when you define the **Logon User** using the service control manager. The *SetISMUser.ps1* file (located at *\ISM\Scripts*) can be used to set the ISM service account to the domain administrator account.
- With a domain user account, the user is not a member of the local administrators. However, the *AddUserRight.ps1* file (located at *\ISM\Scripts*) can be used to grant the "Logon as a service" privilege for the specified user.

In all cases, the ISM account must be defined.

To define the ISM Service:

1. Go to **Control Panel** and start **Services**.
2. Highlight the **Interface Service Manager** service.
3. Right-click and select **Properties** from the context menu.
4. If necessary, enable the service by selecting the startup type of **Manual** or **Automatic**.
5. Select the **Log On** page.
6. Click **This Account**.
7. Select the account you want to use and enter the passwords.

   See the "*Configuring ISM for a Domain User Account* (on page 12)" for more information.
8. Click **OK** to save the startup information and return to the **Services** dialog box.

9. Once you have defined the startup for the service, click **Start**.
10. Use ISM Administrator to configure your service to run under ISM.

Configuring ISM for a Domain User Account

Use the following instructions to configure ISM to run using a domain user account.

To configure ISM to run as a domain user account:

1. Create a domain user account by contacting your domain administrator.
2. Grant the user access to EventLog.
   a. Using Regedit, navigate to *HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\EventLog*.
   b. Select **Permissions**, and grant **Full Control** to the user.
   c. Click **Advanced**.
   d. Select the user, and click **Edit**.
   e. Check **Only apply these permissions to objects/containers in this container**.
   f. Click **OK** to dismiss the registry editor.
3. Grant full access the following folders:
   - The ISM log folder, which is *C:\temp\ISM* by default.
   - The *\ProgramData\Intergraph\Cad\ISM* folder, which is used to manipulate both the ISM service configuration and the application configurations.
   - Any application-specific log file folders. These folders are typically specified using ISM Administrator. Full access is recommended because various logging schemes may need to delete certain files.

   For each folder, use the following steps, substituting <targetFolder> with the folder name.

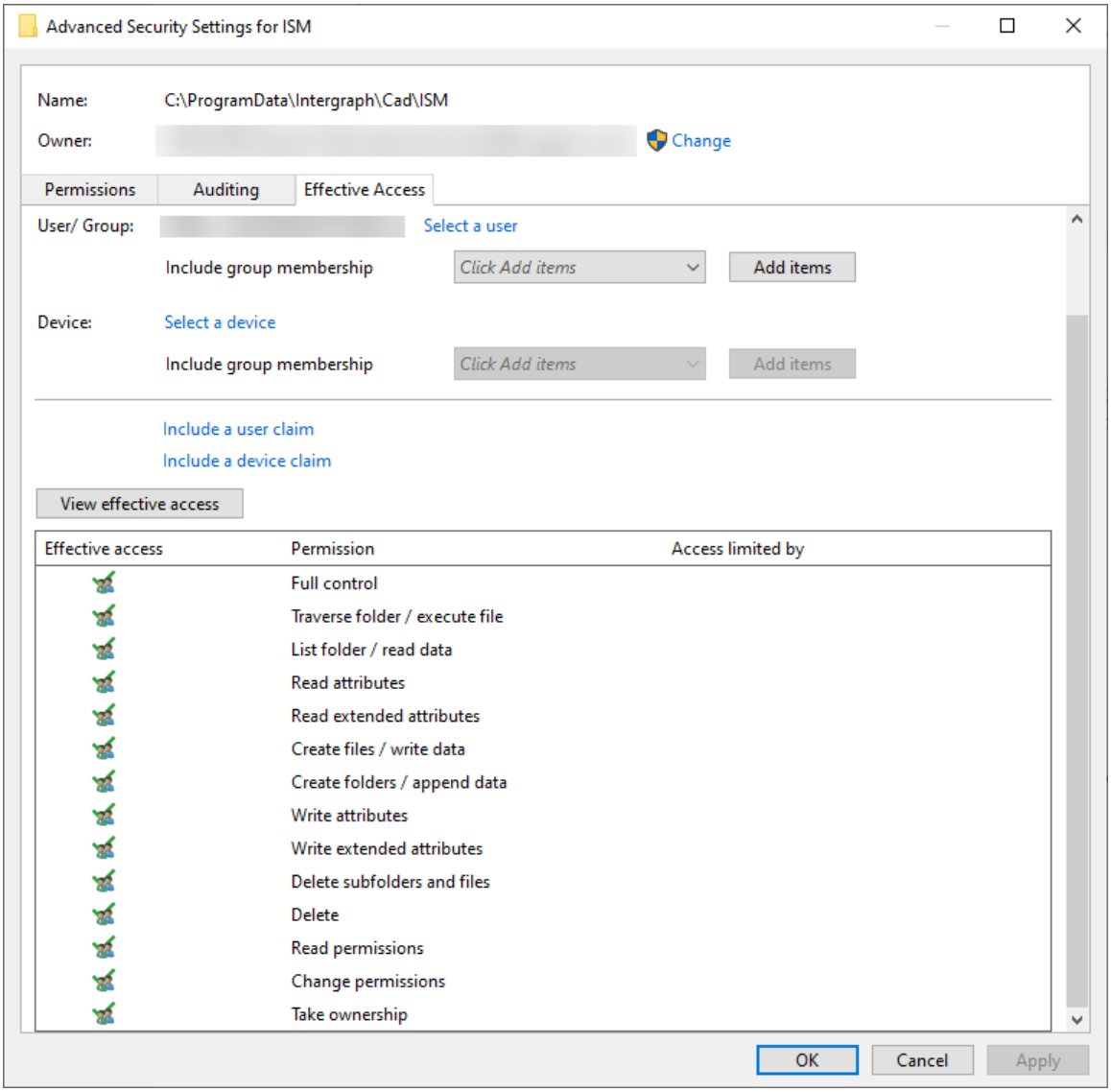
   a. Using Windows Explorer, navigate to *\<targetFolder>*.
   b. Open **Properties** on the ISM folder.
   c. Select the **Security** tab > **Advanced**.
   d. Add the user with **Full Control** permissions to the ISM folder.
   e. Check **Only apply these permissions to objects/containers in this container**.

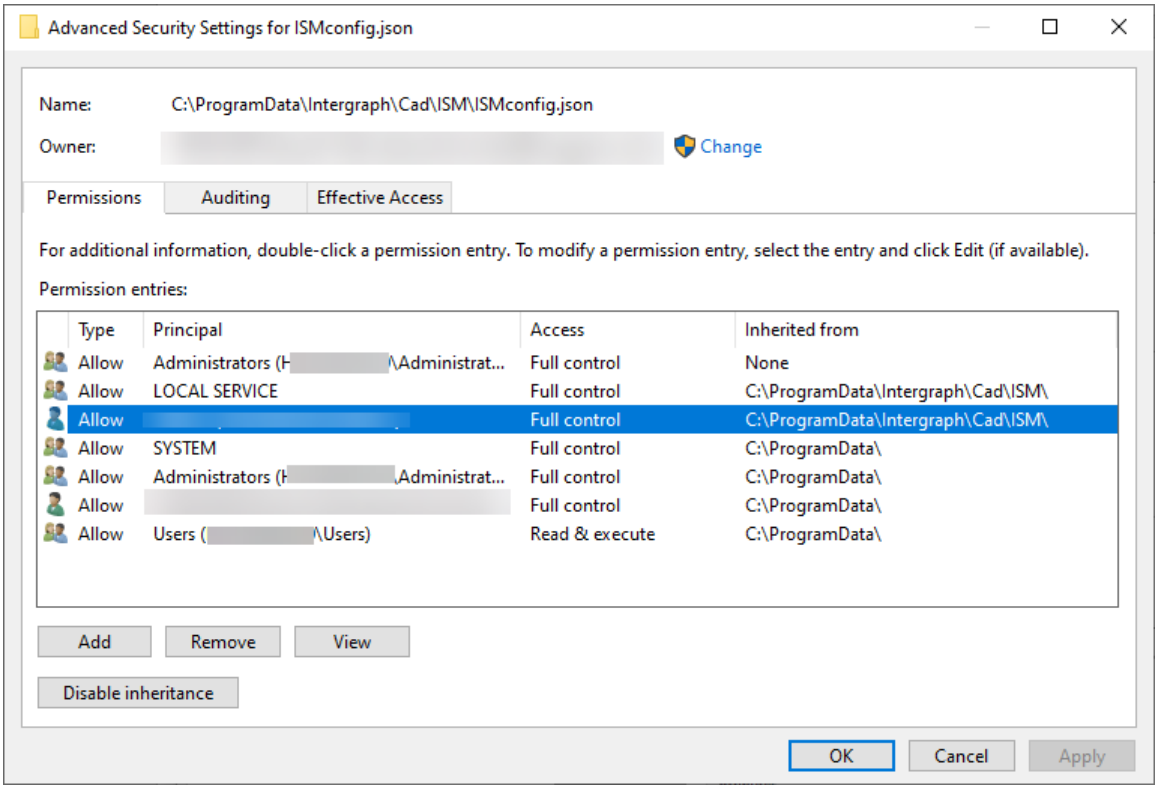
If you encountered a **Failed to enumerate objects in the container. Access is denied** error, run the following commands as an administrator:

```
takeown /f "<drive>\ProgramData\Intergraph\Cad\ISM"
takeown /f "<drive>\ProgramData\Intergraph\Cad\ISM" /r /d y
icacls <drive>\ProgramData\Intergraph\Cad\ISM /grant
Administrators:F
icacls <drive>\ProgramData\Intergraph\Cad\ISM /grant
Administrators:F /t
```

Verify the security is correct on the ISM folder and all files in it. The files should be inheriting the Access Control List (ACL) for the user from the \ISM folder. If the security is not correct, ISM cannot write changes to application configurations, the service configuration, or the ISM history.

For example, the following are the **Expected Effective** access levels for the *\ProgramData\Intergraph\Cad\ISM* folder and the *\ISM\ISMconfig.json* file, respectively.





4. Reserve application HTTP listener ports.

   HTTP listener ports may need to be reserved for one or more of the applications running under ISM. These ports can be reserved using the `netsh http add urlacl` command line. You can run the `netsh` command directly, or run the *ReserveURLNamespace.bat* file delivered with OnCall.CoreServices, which is located by default in *C:\Program Files\Intergraph\OnCall\Core\ConfigurationData\Scripts*.

Service Name	Registration URL
DataServiceModule	http://hostname.mynet.com:9000/
CadBulkLoaderService	http://hostname.mynet.com:9004/
AdminCommandsModule	http://hostname.mynet.com:9010/
ConfigServer	http://hostname.mynet.com:9011/

UnitEventModule	http://hostname.mynet.com:9012/
InformerModule	http://hostname.mynet.com:9020/
InquiryModule	http://hostname.mynet.com:9014/
Automatic Dispatch Service	http://hostname.mynet.com:9037/
RestAPI	https://+:8080/
CadNotificationService	https://+:8081/

Example netsh Commands

- DataServiceModule - `netsh http add urlacl url=http://mycomputer.mynet.com:9000/ user=DOMAIN\user`
- CadBulkLoaderService - `netsh http add urlacl url=http://mycomputer.mynet.com:9004/ user=DOMAIN\user`
- RestAPI - `netsh https add urlacl url= https://+:8080/ user=DOMAIN\user`
- CadNotificationService - `netsh https add urlacl url= https://+:8081/ user=DOMAIN\user`

5. Modify the ISM service to run using the ISM service account:
   a. Open the **Services** applet.
   b. Locate and modify the ISM service **Log On** settings.
   c. Select **This account**. The **Log On As Service** right is added to the account.

If service accounts are used for the system, the service account must have write permission to *C:\Program Files\Intergraph\OnCall.Website\StaticFiles*. Icons on the map do not display if this is not done.

Startup and Shutdown

Once ISM has been properly installed and configured, it acts as a normal Microsoft Windows service and starts automatically upon system boot.

On startup, each of the processes defined through the registry are spawned in the order specified by their StartOrdinal. This information is logged in the application log and the event log and can be viewed using the Event Viewer.

On **Service Shutdown** or **Stop**, the processes are stopped in reverse order based on the defined **Start Sequence**.

RabbitMQ Shutdown

If ISM cannot access RabbitMQ, ISM attempts to connect for 30 seconds before exiting and updating the service status to **stopped** in the Service Control Manager. This can occur during initial startup or losing RabbitMQ after or during initialization.

If ISM loses access to RabbitMQ and cannot reconnect after 30 seconds, the child processes (applications) running under ISM are not forcibly shut down; instead, the child applications determine the parent ISM has exited, and then the child applications should exit on their own. The ISM log contains an entry with **Forcing an exit** just before process termination.

ISM Application Grouping

Application grouping lets ISM handle a group of services as one entity when a group member aborts or fails. As such, application grouping should ease the use of a shared resource such as the local file system as an IPC mechanism.

ISM Administrator Localization

The language used to run ISM Administrator can be configured to meet your site's needs. When starting ISM Administrator, a primary language is loaded initially, and a secondary language can be loaded from the application using the **View > Alternate Language** option.

To enable language configuration, ISM Administrator delivers the following localization files in the *..\Intergraph\ISMAdmin* folder:

- *LocaleMetaData.config*
- Other locale files, for example, *locale-en-us.json* and *locale-en-fr.json*

All localization files should use UTF-8 character encoding to ensure proper functionality.

Using the LocaleMetaData.config File

The *LocaleMetaData.config* file defines a primary language (`firstLanguage`) and optionally defines a secondary language (`alternateLanguage`):

```
{
```

```
  "version": "1.0",
  "sanitizeValueStrategy": "escaped",
  "firstLanguage": {
    "filename": "locale-en-us.json",
    "rtl": "false"
  },
  "alternateLanguage": {
    "filename": "locale-en-fr.json",
    "rtl": "false"
  }
}
```

The following tags are also defined in the *LocaleMetaData.config* file:

- `filename` - references files that contain the localized strings.
- `rtl` (right-to-left) - indicates if the language uses right-to-left script.
- `version` - used to ensure the format of the *LocaleMetaData.config* file is compatible with the running version of ISM Administrator.
- `sanitizeValueStrategy` - used by the underlying translation library, angular-translate. See the online documentation for angular-translate for more information.

Using the Locale Files

The locale (JSON) files consist of tags and localized strings.

Some strings may contain tokens that get replaced by data at runtime.

The following example shows a portion of the *locale-en-us.json* followed by the corresponding portion of the *locale-en-fr.json file*:

English:

```
  "MODIFY_ERR_NAMEREQUIRED": "Name is required",
  "MODIFY_MODIFY": "Modify {{appName}}",
  "MODIFY_GROUPNAME": "Group Name",
```

French:

```
  "MODIFY_ERR_NAMEREQUIRED": "Le nom est requis",
```

```
"MODIFY_MODIFY": "Modifier {{appName}}",
"MODIFY_GROUPNAME": "Nom de groupe",
```

In this example, `{{appName}}` is a token that is substituted at runtime. Other languages can be used by creating appropriate locale files and referencing them from the *LocaleMetaData.config* file.

Configuring Default Preferences

Default user preferences for ISM Administrator, such as the color theme and log folder, can be set using the *\AppData\Roaming\ISMAdmin\preferences.config* file. In particular, the following settings are available:

Setting	Description	Default Value
alternateLanguage	Set to true to enable the alternate language defined in the *LocaleMetadata.config file* (see "*ISM Administrator Localization*" on page 17).	false
darkTheme	Set to true to enable the **Dark Theme** option.	false
diagView	Set to true to display the **Diagnostic View** by default. The diagView and peerView settings should not be set to true at the same time.	false
logFolder	Defines the folder for the ISM Administrator and ISM Connection log files.	*C:\temp\ISMAdmin*
peerView	Set to true to display the **Peer View** by default. The diagView and peerView settings should not be set to true at the same time.	false

These preferences, except for the log folder, can be set using ISM Administrator.

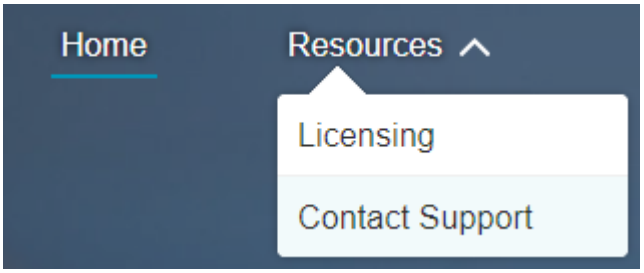
Technical Support

Hexagon's Safety, Infrastructure & Geospatial division provides several ways to contact Support, including online case submission and phone support. For general Hexagon information, call 800-345-4856 (US). For phone support, see the *Contact Support page https://supportsi.hexagon.com/help/s/contactsupport?language=en_US*.

Submit a Case Online

Hexagon provides an online method of requesting support 24/7.

1. Go to the *Support page https://hexagon.com/support-success/safety-infrastructure-geospatial*.
2. Click **Log in**.
3. Type your username and password, and click **Log in**.
4. From the **Resources** list, select **Contact Support**.



   The **Contact Support** page displays.

5. Click **Online case submission**.